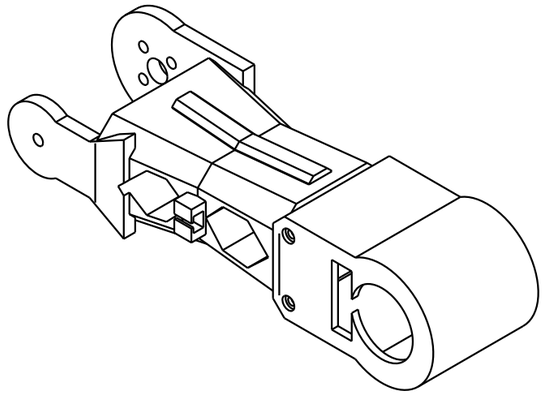
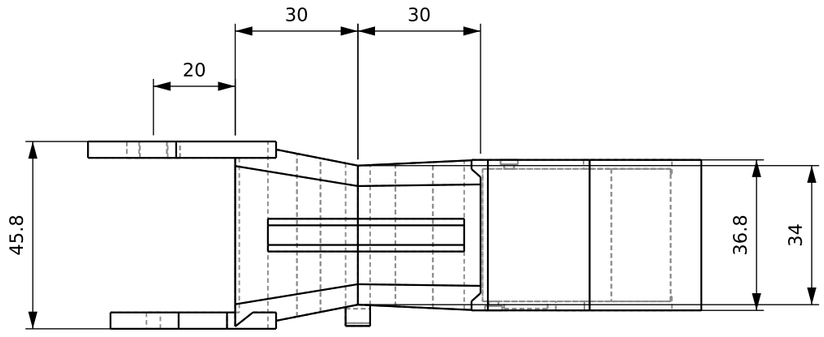
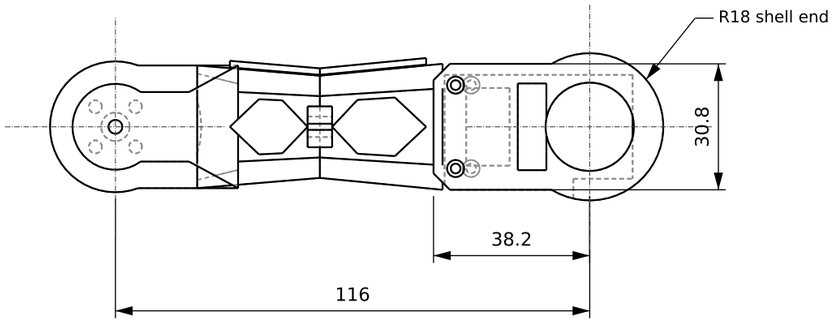


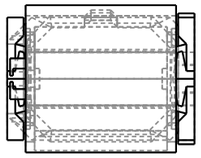
ISOMETRIC (NOT TO SCALE)



TOP



FRONT



RIGHT

FEATURES (in order)

1. Clevis (TAL-002), reach 30, half-height 15.
2. Beam: ruled loft of YZ sections, x 20: 45.8 wide x 30 tall, corners chamfered 6; x 50: 34 wide x 25 tall, corners chamfered 5; x 80: 36.8 wide x 30.8 tall, corners chamfered 6.
3. Servo shell: XZ polygon (81.8,-15.4) (116,-15.4) (116,15.4) (81.8,15.4) (77.8,11.4) (77.8,-11.4), extruded y -18.4..18.4; plus a cylinder R18 about Y at x = 116, y -18.4..18.4.
4. Spine: ruled loft of the ridge profile through the same three stations, then trimmed to x 28..76. Ridge profile at beam height h: (-4,h/2-3) (4,h/2-3) (4,h/2+0.2) (2.4,h/2+1.8) (-2.4,h/2+1.8) (-4,h/2+0.2).
5. Window 1, cut through Y: (28,0) (35.1,6.75) (40.85,6.46) (47,0) (40.85,-6.46) (35.1,-6.75).
6. Window 2, cut through Y: (53,0) (59.19,6.58) (68.35,7.16) (76,0) (68.35,-7.16) (59.19,-6.58).
7. Cable clip (TAL-002) on the -Y flank at the waist: centre (50, -17.2, 0), cable along X.
8. Clevis gap cut (TAL-002).
9. Servo pocket P (sheet TAL-001) with the servo frame origin at (116, 0, 0); servo +Z (shaft) -> +Y, servo +X -> +X, servo +Y -> -Z.
10. Servo opening (drops in from below): cut box x 80.5..112, y -16.2..16.2, z -40..-12.
11. Case screw holes (sheet TAL-002): seat 17.3; shaft face (+Y) at (x, z) = (87, +/-10.2); connector face (-Y) at (83.2, +/-10.2). Each is a dia 2.4 hole from the servo face to the seat, then a dia 4.2 counterbore from the seat outward through the part.

NOTES

- All dimensions mm. Third-angle projection. Do not scale the drawing.
- Part frame = the frame the part is built in, in talos_arm.py (given on each sheet).
- Printed tolerance +/-0.2 unless noted; servo pocket clearances are already included.
- The elbow servo's bus plugs come out through the -Y shell wall.

WORK SET A: base, shoulder, upper arm, hub caps

THIRD ANGLE

PROJECT TALOS 5-axis arm	EST. MASS 51 g
PART **Upper arm**	SCALE 2:3
PART NO. **TAL-103**	UNITS / REV mm / A
QTY / ARM 1	DATE 2026-11-03
MATERIAL PLA, 20 % infill, 3 perimeters	SHEET 6 of 12